Weighted-Proj quadratic-support local ring

2026-10-28 · proof-boundary observation record · no new target measurements

The checked `WeightedProjQuadraticPoint` module constructs the actual weighted-Proj image of an irreducible support satisfying `u | v^2 - f` and proves its affine chart ideal maximal. The new `WeightedProjQuadraticLocal` module proves that its projective stalk is Noetherian and, when the quintic is separable, a domain. The cotangent-rank-one hypothesis has not been proved.

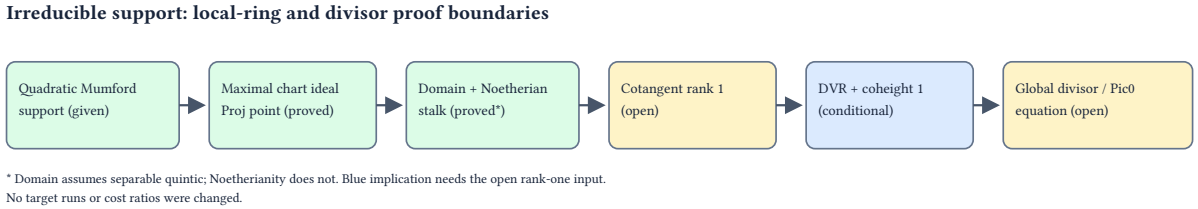

Figure 1: Quadratic-support path. Green statements compile; yellow remains open; blue requires cotangent rank one.

Pinned Mathlib's DVR criterion turns cotangent dimension one into a DVR at this local ring; `WeightedProjCoheight` then gives coheight one. The cotangent dimension, a global closed-point theorem, principal divisor, and Pic0 class equation remain separate. No target data or cost ratio changed.

An unproved next route is to identify the evaluation kernel with the ideal generated by `u(x)` and `y-v(x)`. At nonbranch points, `(y-v)(y+v)=f-v^2` suggests `u(x)` as a local parameter when `y+v` is a unit. The branch case requires its own parameter argument. These are proof obligations, not compiled results.

Sources: Mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`, `AlgebraicGeometry/FunctionField.lean`, `AlgebraicGeometry/Noetherian.lean`, and `RingTheory/DiscreteValuationRing/TFAE.lean`; Stacks Tags 01HD and 00PD; repository `WeightedProjIsIntegral.lean` and `WeightedProjLocallyNoetherian.lean`.

The full cohort cost graphs, rows, and ratios are in `research/four_followups_review_20261022/`. This report changes none of those results.